

Short-Horizon Pedestrian Forecast from Onboard Hip Orientation

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Abstract—A robot that treats a nearby walker as a fixed blob on the costmap will plan around where the person is now. Prior work on the freezing-robot problem, on pose-aware trajectory models, and on constant-velocity baselines all point at a shorter question: does a body heading measured on the robot improve a one-second forecast enough to be worth keeping? This paper answers that with an onboard stereo body tracker and an extended Kalman filter. Each update uses the same state $[p_x, p_y, v, \theta, \omega]$ and the same hip heading from ZED BODY_18. The filter then rolls that posterior forward for 1s under two models. CTRV turns with the estimated rate ω . Hip-steering turns heading toward the measured hip angle with a fixed gain and then steps forward at speed v . We replay two walks, a D-shaped path and a straight path. Only the motion model changes. Mean 1s displacement error on the D-walk is 0.192m for hip-steering and 0.203m for CTRV (537 samples). On the straight walk the means are 0.198m and 0.206m (232 samples). Final-step error is lower for hip-steering on the straight walk and slightly higher on the D-walk. A live predicted pose is shown in RViz. We do not report a closed-loop Nav2 timing study.

I. INTRODUCTION

Local planners that read a costmap usually mark a nearby walker at the pose the sensor reports now. If that cell sits on the path, the robot stops or detours around a person who is already leaving. Trautman and Krause named the stall that follows the freezing-robot problem [1]. Frozone and later crowd-navigation papers treat the same timing from the planner side: a robot that will pass someone in about a second needs a short forecast [2].

Body heading is one cheap cue for that forecast. Unhelkar et al. used anticipatory walking signals for co-navigation [3]. Quintero et al. used pose and activity for path intention [4]. Salzmann et al. used human pose for trajectory prediction [5]. Schöller et al. showed that a constant-velocity model is often hard to beat on pedestrian data [6]. Those papers leave hip heading as an external cue or as a learned pose feature. We put the hip angle inside the motion model of a small extended Kalman filter and score a one-second open-loop rollout against a constant-turn-rate-and-velocity (CTRV) model on the same onboard tracks.

A ZED camera runs BODY_18 tracking. Each update measures planar position and a hip heading from the left and right hip keypoints. The filter state is $[p_x, p_y, v, \theta, \omega]$. After the update, the posterior is stepped for 1s. CTRV turns at the estimated rate ω . Hip-steering turns heading toward the measured hip angle with a fixed gain $b = 3.0$, then moves at speed v . Two walks are replayed, a D-shaped path and a straight path. Only the motion model changes.

Mean 1s displacement error on the D-walk is 0.192m for hip-steering and 0.203m for CTRV (537 samples). On the straight walk the means are 0.198m and 0.206m (232 samples). Final displacement error is lower for hip-steering

on the straight walk and slightly higher on the D-walk. The live predicted pose is shown in RViz. We do not report closed-loop Nav2 metrics.

II. RELATED WORK

Trautman and Krause described the freezing-robot problem: if other people are modeled as uncertain moving obstacles, a planner can decide that every forward path is unsafe [1]. They argued that better single-agent prediction by itself does not remove the stall once the crowd is dense enough. Frozone tracks pedestrians, builds a predicted freezing zone, and steers a real robot around it [2]. Lu et al. split the ROS costmap into layers so different constraints can be written separately [7]. Katyal et al. attach intent-aware prediction to an adaptive navigation policy and report collisions and personal-space violations, not only displacement error [8].

Schöller et al. showed that a constant-velocity model can match or beat several neural predictors on standard pedestrian benchmarks [6]. Sun et al. learn 3DOF pose trajectories (planar position plus heading) from long-term range-finder deployments with an LSTM [9]. González et al. replace a basic kinematic step with a short-term gait-biomechanics predictor meant to run on a small device [10].

Walking turn cues show up before the path bends. Unhelkar et al. measured those cues and used them as planner features [3]. Quintero et al. map body keypoints into Gaussian-process models and predict path, pose, and activity about 1s ahead for vehicles [4]. Onboard cameras add more of the skeleton: Salzmann et al. feed positions, head orientation, and 3D keypoints into a Transformer [5], and Nilavadi et al. predict full-body pose dynamics together with the root trajectory [11].

This paper keeps a five-state EKF and changes only how heading is advanced. CTRV uses the estimated rate ω . Hip-steering uses BODY_18 hip heading with a fixed gain. The score is a 1s open-loop position error on two SVO walks.

III. METHOD

A ZED stereo camera runs BODY_18 tracking. Filter position uses the camera horizontal plane: p_x is the body Z coordinate (forward) and p_y is the body X coordinate (lateral). Hip heading uses keypoints 8 (right hip) and 11 (left hip). The hip line is taken in that plane, and body forward is its perpendicular. If ground speed is above 0.05m/s and that perpendicular points against the velocity, the heading is flipped. If a hip is missing or the hip line is shorter than 10^{-6} m, the heading is the velocity angle.

The filter state and measurement are

$$\mathbf{x} = [p_x, p_y, v, \theta, \omega]^\top, \quad \mathbf{z} = [p_x, p_y, \theta_{\text{hip}}]^\top. \quad (1)$$

The predicted measurement is $h(\mathbf{x}) = [p_x, p_y, \theta]^\top$, so the heading residual is the wrapped difference $\theta_{\text{hip}} - \theta$. Each camera frame runs one EKF predict with the chosen motion model, then one update with $\mathbf{z}$. Frame Δt comes from the camera timestamps. The initial covariance, process noise, and measurement noise are fixed for both models: $P_0 = \text{diag}(1, 1, 100, 10, 5)$, $Q = \text{diag}(0.01, 0.01, 0.1, 0.05, 0.01)$, $R = \text{diag}(0.1, 0.1, 1.0)$.

A. Motion models

CTRV keeps speed v and turn rate ω and advances heading by $\omega\Delta t$. When $|\omega| > 10^{-5}$, the EKF predict uses the constant-turn arc

$$\begin{aligned} p_x &\leftarrow p_x + (v/\omega)(\sin(\theta + \omega\Delta t) - \sin\theta), \\ p_y &\leftarrow p_y + (v/\omega)(-\cos(\theta + \omega\Delta t) + \cos\theta). \end{aligned} \quad (2)$$

Otherwise it steps straight at the current heading.

Hip-steering leaves v unchanged and turns heading toward the current hip angle,

$$\theta \leftarrow \theta + b \sin(\theta_{\text{hip}} - \theta) \Delta t, \quad (3)$$

then steps

$$p_x \leftarrow p_x + v \cos \theta \Delta t, \quad p_y \leftarrow p_y + v \sin \theta \Delta t. \quad (4)$$

The stored ω is that heading change divided by Δt . The paper runs use $b = 3.0$. Hip heading is held fixed during one predict step.

B. One-second forecast

After the update, the posterior is copied and integrated for $T = 1.0$ s with no new measurements, in 20 equal steps. CTRV turns at the posterior ω and steps with v . Hip-steering keeps applying (3) toward the last θ_{hip} , then steps with v . That last hip angle does not change over the 20 steps. The scored point is the position at T .

IV. EXPERIMENTAL SETUP

Both models are run on the same two SVO replays, a D-shaped walk and a straight walk. The EKF, measurements, covariances, and horizon stay the same. Only the motion model changes.

A frame is scored when a later measurement exists at the target time. The error is the Euclidean distance in the (p_x, p_y) plane between that measurement and the 1 s forecast. ADE is the mean of those errors. What we report as FDE is the 1 s error on the last scored frame of the log. Max and median use the same list. The D-walk has 537 scored frames. The straight walk has 232.

V. RESULTS

In Table I, hip-steering ADE is 0.011 m lower on the D-walk and 0.008 m lower on the straight walk. FDE is 0.026 m higher on the D-walk and 0.018 m lower on the straight walk. The max errors differ by a few millimetres. Median error is higher for hip-steering on the D-walk (0.175 m against 0.157 m) and equal on the straight walk. Fig. ?? is the D-walk forecast plot.

TABLE I
1 S FORECAST ERROR. FDE IS THE LAST SCORED FRAME.

Scene	Model	ADE (m)	FDE (m)	Max (m)	Median (m)	N
D-walk	CTRV	0.203	0.502	0.843	0.157	537
D-walk	Hip-steering	0.192	0.528	0.844	0.175	537
Straight	CTRV	0.206	0.111	0.655	0.153	232
Straight	Hip-steering	0.198	0.093	0.649	0.153	232

D-walk 1 s forecast (hip vs CTRV).
Placeholder. Swap in the PNG.

Fig. 1. D-walk 1 s open-loop forecast. Replace this box with the evaluation plot.

VI. NAVIGATION INTEGRATION AND LIMITS

A live node draws the current person pose and the 1 s forecast in RViz on `/person/danger_zone`. That topic is a marker array. The Nav2 local costmap we run still inflates laser hits. There is no predicted-person layer, and there is no corridor trial with success rate or time stuck. Table I also has no static baseline that copies the current position forward. FDE is the 1 s error on the last scored frame of each log, not a mean over terminal horizon steps.

VII. CONCLUSION

On the D-walk, 1 s ADE is 0.192 m for hip-steering and 0.203 m for CTRV (537 frames). On the straight walk the means are 0.198 m and 0.206 m (232 frames). FDE is lower for hip-steering on the straight walk and higher on the D-walk. A closed-loop Nav2 comparison is not in this paper.

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